Yang Sen (Liam) Lin

Personal Statement: Columbia MSCS (VGIR Track)

Program: Master of Science in Computer Science (MSCS)

I aspire to ***architect trustworthy embodied intelligence*** *by unifying the semantic reasoning of vision-language models with the provable safety guarantees of control theory.*

In my recent research accepted to **CoRL 2025**, titled "**See, Point, Fly**," I identified a fundamental trade-off in the current state of robotic autonomy: geometric rules provide reliability but become brittle in dynamic environments; conversely, Generative Vision-Language-Action (VLA) models offer adaptability but suffer from hallucinations and high computational latency. My objective at **Columbia University** is to resolve this tension. I aim to architect systems where the semantic creativity of VLMs is grounded by physical constraints. This is not merely an algorithmic challenge; it requires a convergence of **Structured World Models and Robust Control**—a synthesis that specifically demands the academic resources at Columbia.

My journey toward this objective began with a focus on algorithmic decoupling. In my **CoRL 2025** work, I formulated a geometric projection algorithm to ground 2D VLM prompts into 3D metric controls, achieving a **93.9% success rate** in unstructured environments. However, rigorous testing revealed a "**reactivity bottleneck**": my static geometric priors failed against dynamic, erratic obstacles. This failure highlighted a critical insight: while geometry provides structure, it lacks dynamics. To evolve from static heuristics to adaptive policies, I must incorporate physics directly into the learning loop. This specific technical impasse drives my decision to apply to Columbia. I specifically need the expertise of **Prof. Yunzhu Li** to elevate my understanding of structured representations into physics-informed world models, and the guidance of **Prof. Matei Ciocarlie** to ensure these learning-based policies adhere to provable safety dynamics in the real world.

Beyond algorithms, my research philosophy is defined by the constraints of the physical layer. In safety-critical robotics, inference latency is not just a performance metric; it is a failure mode. My internship at **Wolley Inc.** grounded me in this reality. There, I engineered C-based diagnostic tools to profile **CXL memory bandwidth**. This experience taught me that algorithmic safety is irrelevant if the hardware stack cannot support the control frequency. Recognizing this "full-stack" bottleneck, I was selected as an incoming **Research Intern at OMRON SINIC X**. This opportunity validates my potential to bridge the gap between high-level reasoning and edge deployment. I see the **Vision, Graphics, Interaction, and Robotics (VGIR)** track at Columbia as the essential academic bridge that connects my hardware profiling experience with high-level AI reasoning, transforming me into a full-stack **Computational Systems Architect**.

My academic preparation is defined by the convergence of **Mechanical Engineering and Computer Science**. While ME grounded me in the physical first principles of system dynamics, my dual focus in CS equipped me with the algorithmic reasoning of modern AI. I demonstrated the ability to synthesize these fields at a graduate level by securing an **A+ in Graduate Computer Vision** and applying Projective Geometry to derive non-linear unprojection laws in my CoRL paper. This practical application prepares me for the rigorous theoretical training of the MSCS program. I specifically plan to take **EEME E6602 Modern Control Theory** to formalize heuristic safety constraints into provably stable guarantees, and **COMS E6998 Topics in Robot Learning** to master the data-driven techniques necessary to move beyond my current zero-shot frameworks. Furthermore, to support my goal of real-time deployment, I look forward to **EECS E4750 Heterogeneous Computing** to explore hardware acceleration for heavy foundation models.

Finally, the "**Engineering for Humanity**" mission of Columbia Engineering resonates with my desire to build technology that functions reliably in the real world. As the **GDSC Chapter Lead**, I bridged the gap between theory and practice for over 40 student developers, and I am eager to bring this collaborative spirit to the NYC tech ecosystem. The **Columbia MSCS** is the pivotal step that transforms me from a robotics engineer who builds working prototypes into a **Computational Systems Architect** capable of defining the safety guarantees required to deploy autonomous agents at a global scale.